

as taught in the work shop during the previous classes we install numpy

we import and call it out in the code
import numpy as np

According to the research i took on w3 schools below is the Function to validate input as a number
def validate_input(prompt):
 while True:

try:
 value = float(input(prompt))
 return value
 except ValueError:
 print("Invalid input. Please enter a valid number.")

further more i used the function create and range to input the value of the matrix a 3x3 matrix
from user input and print it out in a matrix form

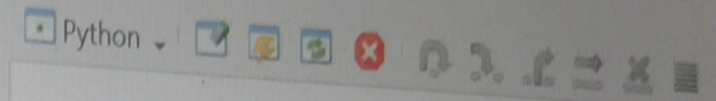
def create_matrix():
 matrix = []
 for i in range(3):
 row = []
 for j in range(3):
 value = validate_input(f"Input the number of the matrix for A[{i+1}][{j+1}]: ")
 row.append(value)
 matrix.append(row)
 return matrix

print("Enter values for a 3x3 Matrix:")
matrix_A = create_matrix()
A = np.array(matrix_A)
print("\nMatrix A:")
print(A)

Further more calculating the determinant of the matrix
determinant = np.linalg.det(A)
print("\nDeterminant of A:", determinant)

to check if matrix A, determinant is = 0
if determinant != 0:

Calculate the inverse of the matrix



```
>>> (executing file "PYTHON question 1.py")  
Enter values for a 3x3 Matrix:  
Input the number of the matrix for A[1][1]: 1  
Input the number of the matrix for A[1][2]: 2  
Input the number of the matrix for A[1][3]: 3  
Input the number of the matrix for A[2][1]: 9  
Input the number of the matrix for A[2][2]: 8  
Input the number of the matrix for A[2][3]: 7  
Input the number of the matrix for A[3][1]: 6  
Input the number of the matrix for A[3][2]: 5  
Input the number of the matrix for A[3][3]: 4  
Invalid input. Please enter a valid number.  
Input the number of the matrix for A[3][3]: 4
```

Matrix A:
[[1. 2. 3.]
 [9. 8. 7.]
 [6. 5. 4.]]

Determinant of A: 0.0

Source structure

validate_input()
create_matrix()

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```

        value = float(input(prompt))
        return value
    except ValueError:
        print("Invalid input. Please enter a valid number.")

```

further more i used the function create and range to input the valuse of the matrix a 3x3 matrix from user input and print it out in a matrix form

```

def create_matrix():
    matrix = []
    for i in range(3):
        row = []
        for j in range(3):
            value = validate_input(f"Input the number of the matrix for A[{i+1}][{j+1}]: ")
            row.append(value)
        matrix.append(row)
    return matrix
print("Enter values for a 3x3 Matrix:")
matrix_A = create_matrix()
A = np.array(matrix_A)
print("\nMatrix A:")
print(A)

```

Further more icalculating the determinant of the matrix

```

determinant = np.linalg.det(A)
print("\nDeterminant of A:", determinant)

```

to check if matrix A,determinant is = 0

```

if determinant != 0:
    # Calculate the inverse of the matrix
    inverse_A = np.linalg.inv(A)
    print("\nInverse of A:")
    print(inverse_A)

```

the program calculates $A * A$ and $A * A * A$ using matrix multiplication and prints the results.

```

A_squared = np.dot(A, A)
A_cubed = np.dot(A_squared, A)
print("\nA * A:")
print(A_squared)
print("\nA * A * A:")
print(A_cubed)
count = np.sum(A >= 10)
print("\nNumber of values in A >= 10:" count)

```



```
#Name: Oladipupo Roland Familua  
#Student Id. 2326310
```

```
import numpy as np  
#According to my research the function or operator called trace takes a square matrix as input and uses  
the np.trace function from numpy to calculate the trace of the matrix, which is the sum of the elements  
on the main diagonal. The result is then returned it is usually denoted as Calculate_Trace  
def calculate_trace(matrix):  
    trace = np.trace(matrix)  
    return trace
```

```
#below is to find the trace for 3x3 matrix  
matrix_3x3 = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])  
trace_3x3 = calculate_trace(matrix_3x3)  
print("Trace of 3x3 matrix:", trace_3x3)
```

```
# below is to find the trace for 4x4 matrix  
matrix_4x4 = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 16]])  
trace_4x4 = calculate_trace(matrix_4x4)  
print("Trace of 4x4 matrix:", trace_4x4)
```

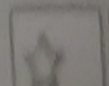
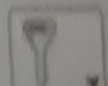
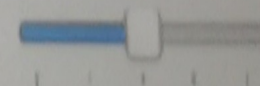
```
below is to find the trace for 6x6 matrix  
matrix_6x6 = np.array([[1, 2, 3, 4, 5, 6], [7, 8, 9, 10, 11, 12], [13, 14, 15, 16, 17, 18],  
                        [19, 20, 21, 22, 23, 24], [25, 26, 27, 28, 29, 30], [31, 32, 33, 34, 35, 36]])  
trace_6x6 = calculate_trace(matrix_6x6)  
print("Trace of 6x6 matrix:", trace_6x6)
```

```
>>> (executing file "PYTHON question 2.py")  
Trace of 3x3 matrix: 15  
Trace of 4x4 matrix: 34  
Trace of 6x6 matrix: 111
```

```
>>>
```

Source structure

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Click star


```
#Name; Oladipupo Roland Familua
```

```
#Student ID; 2326310
```

```
def ackermann(m: int, n: int) -> int:
    #following are Ackermann's function.
    #m as int is the First parameter of Ackermann's function.
    #n as int is the Second parameter of Ackermann's function.
    if m == 0:
        return n + 1
    elif m > 0 and n == 0:
        return ackermann(m - 1, 1)
    elif m > 0 and n > 0:
        return ackermann(m - 1, ackermann(m, n - 1))
```

```
# Test the function
```

```
m = 2
n = 3
result = ackermann(m, n)
print(f"A({m}, {n}) = {result}")
```

Shells

Python

```
>>> (executing file "PYTHON question3.py")
A(2, 3) = 9
```

```
>>>
```

Source structure

ackermann()

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